Project: **Molecular Dynamics (MD)** simulation of **solid Ar**

Alternative: Instead of Ar, consider Ne, Kr, Xe…

Goal: Create a MD code for a Lennard-Jones (LJ) system in solid phase. Start from code of practice 2.

Outline:

1. Write a MD code for a (N,V,T) simulation of a Lennard-Jones (LJ) particles. Use N=500 atoms. You'll need to use a Berendsen's thermostat.

2. Create an initial set-up of positions from the sites of a fcc *(face centred cubic*) lattice (HINT: consider a fcc lattice as the superposition of 4 single cubic lattices)

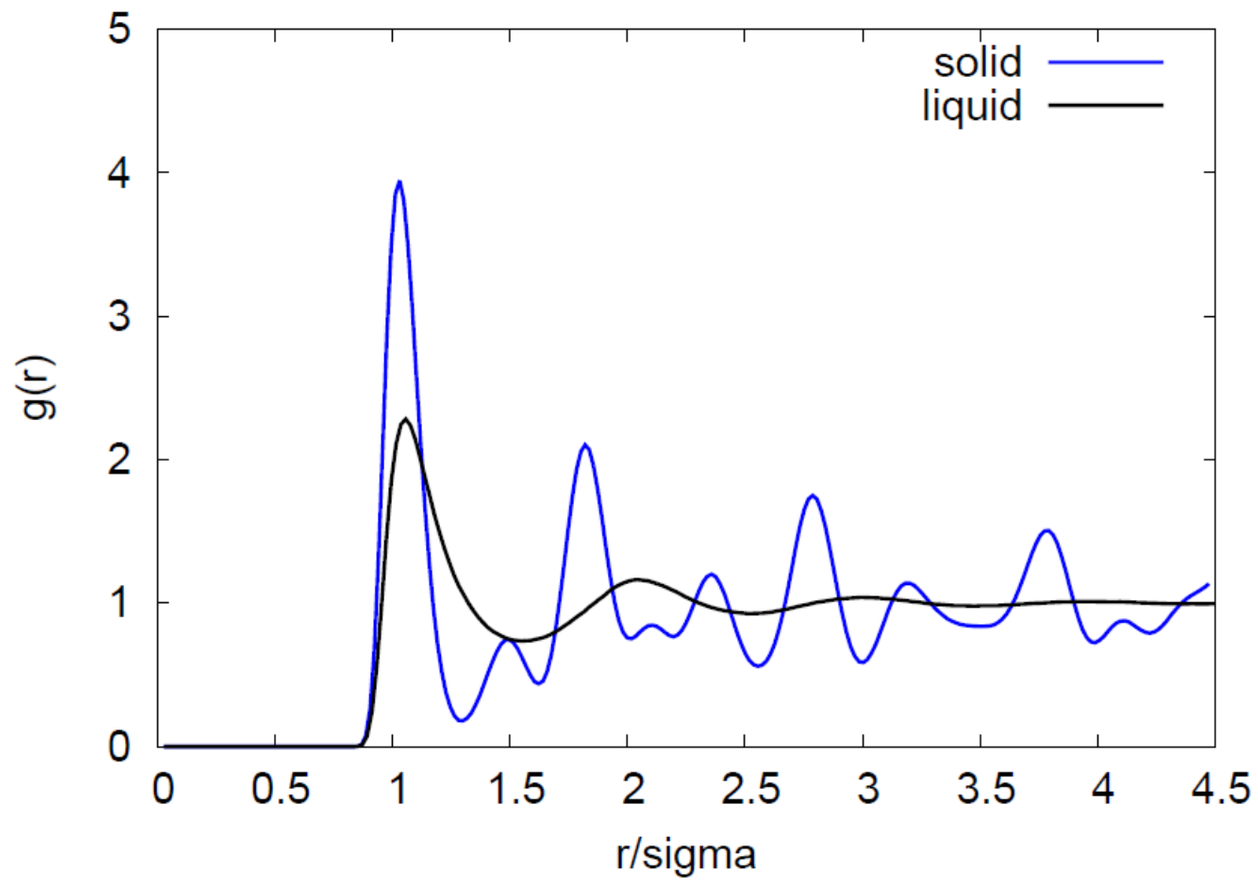
3. Run a MD simulation for argon at T*= 1.2 and at ρ*= 1.2. Compute the energy, the pressure, the radial distribution function of the system, the velocity correlation function and the mean square displacement.

4. Repeat the simulation for the following sate points: T*= 1.2 and ρ*= 0.8 ; T*=2.0 and ρ*= 0.8 ; T*=2.0 and ρ*= 1.2 , and compare with the previous results.

Lennard-Jones interaction potential

atom	mass(uma)	ε/kB (K)	σ(Å)
Ar	39.948	119.8	3.405

Some results to compare with:



Questions:

1. What are the main differences with the liquid argon case? Why?
2. Make Fourier transforms of the VACF and obtain spectral densities. What are the main vibrational motions in the system? How this is different of the liquid system?
3. Derive the correction to the pressure for the LJ potential due to the cutoff of the interaction potential.